
Tarski Hierarchies

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VOLKER HALBACH

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ABSTRACT. The general notions of object- and metalanguage are discussed and as a special case of this relation an arbitrary first order language $\mathcal{L}_0$ with an infinite model is expanded by a predicate symbol T_0 which is interpreted as truth predicate for $\mathcal{L}_0$. Then the expanded language is again augmented by a new truth predicate T_1 for the whole language $\mathcal{L}_0$ plus T_0 . This process is iterated into the transfinite to obtain the Tarskian hierarchy of languages. It is shown that there are natural points for stopping this process. The sets which become definable in suitable hierarchies are investigated, so that the relevance of the Tarskian hierarchy to some subjects of philosophy of mathematics are clarified.

It should be noticed that these terms “object language” and “meta language” have only a relative sense. If, for instance, we become interested in the notion of truth applying to sentences, not of our original object-language, but of our meta-language, the latter becomes automatically the object-language of our discussion; and in order to define truth for this language, we have to go to a new meta-language — so to speak, to a meta-language of a higher level. In this way we arrive at a whole hierarchy of languages.

(Tarski, 1986, p. 674f)

1. HIERARCHIES OF OBJECT- AND METALANGUAGES

Under what circumstances is a formal language a metalanguage for another formal language? — If it is possible to “speak” within the first language about the semantics of the second one, or, more precisely, if the first one contains a truth predicate for the latter. To make this still more precise the notion of a *language* must be fixed. In formal logic a language is often understood to be a set of formulas closed under the usual rules of formation of formulas, a calculus for constructing formulas or something similar. Obviously, this notion of a language does not make sense for the above definition of the relation between object- and metalanguage, because it does not determine, whether a predicate is a truth predicate. So there has

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to be, in addition, either an interpretation or an derivation¹ system for the language. To make both views more explicit we shall assume that two uninterpreted and unaxiomatized first-order languages $\mathcal{L}_O$ and $\mathcal{L}_M$ are given. As there is no interpretation or axiomatization for these languages assumed, “first-order” means here only that the languages contains the usual symbols for the connectives and the quantifier(s) and is closed under the usual rules of formation of formulas. Furthermore, we assume in the present section the existence of a closed term $\bar{\phi}$ of $\mathcal{L}_M$ for every sentence ϕ of $\mathcal{L}_O$. If there is a model specified for $\mathcal{L}_M$ then by stipulation the objects denoted by this terms will be called the Gödel numbers of the respective sentences in this model.

According to the semantical approach, “language” is going to be understood as “interpreted language” in the above definition of “metalanguage” and consequently a model $\mathfrak{M}_M$ for the language $\mathcal{L}_M$ is required. In order to tell whether a formula $\tau(x)$ is a truth predicate for the language $\mathcal{L}_O$, the following conditions might be considered:

- (i) $\tau(x)$ is a truth predicate for $\mathcal{L}_O$, if and only if for all sentences ϕ of $\mathcal{L}_O$ the following equivalence is valid:

$$\mathfrak{M}_M \models \tau(\bar{\phi}), \text{ if and only if } \phi \text{ is true.}$$

For this definition a model for $\mathcal{L}_O$ is required, too, because it is necessary to refer to a model in order to tell whether ϕ is true or not. It is a pleasant and sometimes useful feature of this condition that it can be formulated without the closed terms $\bar{\phi}$ by saying that the extension of T in the model $\mathfrak{M}_M$ should be the set of (Gödel numbers of) all true sentences of $\mathcal{L}_O$.

- (ii) $\tau(x)$ is a truth predicate for $\mathcal{L}_O$, if and only if $\mathfrak{M}_M \models \tau(\bar{\phi}) \leftrightarrow \phi$ holds for all sentences ϕ of $\mathcal{L}_O$. Note that the equivalences $\tau(\bar{\phi}) \leftrightarrow \phi$, usually known as the Tarski biconditionals, are formulated in the language $\mathcal{L}_M$. So $\mathcal{L}_M$ has to contain the language $\mathcal{L}_O$ according to this definition². On the other hand, with this definition no model for $\mathcal{L}_O$ is required.

Instead of the Tarski biconditionals other principles may be considered in (ii), e.g. an axiom saying that the conjunction of two sentences is true, if and only if both sentences from which it is built up are true. But as long as only acceptable models are considered the addition of this and similar principles will not make a difference.

Obviously, (i) implies (ii), i.e. if τ is also a truth predicate in the sense of (i) it is a truth predicate in the sense of (ii), provided that $\mathfrak{M}_M$ is an expansion of $\mathfrak{M}_O$.

If we stick to Tarski’s original syntactic approach, we are facing a similar situation as in the preceding semantical proposal: Instead of validity in the

model $\mathfrak{M}_M$ we are dealing with derivability in a theory S_M for the language $\mathcal{L}_M$. If we perform this substitution in (i), we get the following clause:

- (i') $\tau(x)$ is a truth predicate for $\mathcal{L}_O$, if and only if for all sentences ϕ of $\mathcal{L}_O$ the following holds:

$$S_M \vdash \tau(\overline{\phi}), \text{ if and only if } \phi \text{ is true.}$$

For this definition we need again a model $\mathfrak{M}_O$ for the language $\mathcal{L}_O$ in order to express the right-hand side of the equivalence.

If $\mathcal{L}_O$ is a language of arithmetic and $\mathfrak{M}_O$ is the standard model of arithmetic, S_M cannot be a recursively enumerable theory, rather, it will be exactly Δ_1^1 by the well-known theorem on the complexity of the set of all true sentences of arithmetic. As we cannot have a derivation system for generating such a complex set of sentences, this definition is not satisfactory, because we were forced to associate with $\mathcal{L}_M$ not only a recursively enumerable theory or a corresponding derivation system but a non-arithmetical set in contrast to our original aim. So we turn to the reformulation of condition (ii) in terms of axiomatic systems instead of models. The definition we obtain is very similar to Tarski's.

- (ii') $\tau(x)$ is a truth predicate for $\mathcal{L}_O$, if and only if all Tarski biconditionals $\tau(\overline{\phi}) \leftrightarrow \phi$ for sentences ϕ of $\mathcal{L}_O$ are derivable in S_M .

In contrast to (ii) above, completely different explications of the notion of a metalanguage are obtained, if other (stronger) theories are used in (ii') instead of the set of all Tarski biconditionals. For example, the Tarski biconditionals do not imply the sentence (if it can be expressed at all) saying that a conjunction is true, if and only if both conjuncts are true³. Such stronger theories mirroring the usual Tarskian inductive definition of truth were studied in great detail (see (Halbach, 1994, chap. 7) for further information).

In this paper we shall deal only with the semantical approach. Results on this will form a good guide also for syntactical explications of the object-metalanguage relation and for hierarchies of object-meta-theories, which we shall treat in another paper.

If certain minimal assumptions about the language $\mathcal{L}_M$ and the associated model $\mathfrak{M}_M$ are made, then for $\mathcal{L}_M$ it is not possible to be the metalanguage of itself in the sense of (ii) and hence of (i), provided that "is true" in (i) is understood as "true in $\mathfrak{M}_M$ ". On the other hand, it is quite obvious how to construct a metalanguage in the sense of (i) for a given language $\mathcal{L}_O$ with an associated model $\mathfrak{M}_O$. The simplest and most trivial solution consists in a language $\mathcal{L}_T$ containing only the constants for each sentence ϕ of $\mathcal{L}_O$ and a one-place predicate letter T as its non-logical symbols which has exactly the set of true sentences of $\mathcal{L}_O$ as its extension,

while the constants are interpreted by the corresponding sentences of $\mathcal{L}_O$ or their respective Gödel numbers; the domain of the model is simply the set of all sentences of $\mathcal{L}_O$. Of course, such a metalanguage is not quite satisfactory because of its lack of expressive power. A good and often used remedy against this shortcomings is the requirement that a metalanguage should contain the objectlanguage as a sublanguage and we shall make this assumption in the following. It will us also enable to formulate the Tarski biconditionals within the metalanguage. With these assumptions the minimal metalanguage $\mathcal{L}_M$ for the language $\mathcal{L}_O$ with the model $\mathfrak{M}_M$ is the language $\mathcal{L}_O$ augmented by an additional one-place predicate symbol T , which is going to be interpreted in the new model for the augmented language as in the case of $\mathcal{L}_T$, provided that we disregard all the trivial cases where a language contains its own metalanguage. If the sentences of $\mathcal{L}_O$ or corresponding Gödel numbers do not form already a subset of the domain of $\mathfrak{M}_O$ we have to add them to the domain of the model $\mathfrak{M}_M$. All symbols of $\mathcal{L}_O$ are interpreted in $\mathfrak{M}_M$ as in $\mathfrak{M}_O$, except that it must be fixed how the symbols of $\mathcal{L}_M$ should behave on the new objects, i.e. on the sentences of $\mathcal{L}_O$ which were possibly added to the domain. Here it is difficult to find a plausible uniform solution, because this will depend on the choice of $\mathcal{L}_O$ and the model $\mathfrak{M}_O$. The metalanguage $\mathcal{L}_M$ with the associated model $\mathfrak{M}_M$ generated in this way is a metalanguage in the sense of (i) as well in the sense of (ii) and it is minimal, again according to (i) and (ii), if $\mathcal{L}_O$ is a sublanguage of $\mathcal{L}_M$. If the model $\mathfrak{M}_O$ includes all sentences (or their Gödel numbers) in its domain — and this is the case for all models we shall consider —, this construction of a metalanguage may be performed in a uniform way. Because of this uniformity it is possible to iterate the transition to a metalanguage. In this way we can generate Tarski's hierarchy of object- and metalanguages, which is mentioned in the motto of this paper.

Why should this hierarchy raise interest? In practice, a metalanguage is usually much stronger than the objectlanguage, because we work often within a set theoretical language augmented by additional vocabulary in order to speak about an objectlanguage. So the transition from the object- to the metalanguage is not by any means canonical. But the metalanguage constructed in the above way is under usual conditions contained in such a strong metalanguage. Hence we get information about the minimum expressive power of a hierarchy of object- and metalanguages and related results, if we study this hierarchy consisting of minimal metalanguages. The results of this paper have therefore also some relevance for hierarchies with non-uniform transitions from object- to metalanguage. The iteration

of the transition might be considered as a formalization of an attempt to find always stronger and stronger metalanguages.

Traditionally, the most prominent reason to explore such hierarchies is given by the liar paradox and related topics (see e.g. (Burge, 1979), (Kripke, 1975) and (Parsons, 1974)). In 1975 Kripke complained about the lack of an elaborated theory of the Tarskian hierarchy so that it was at that time impossible to compare alternative approaches to a solution of the liar paradox with the hierarchy. Although there were some substantial contributions to the theory of the hierarchies, e.g. (Feferman, 1991), a systematic treatment is still missing, at least as far as we know.

According to Tarski's theorem on the undefinability of truth the expressive power of a language is increased, if a truth predicate for this language is added. By expanding the language in this way again and again we obtain stronger and stronger languages and more and more sets become definable. The addition of a truth predicate may be seen as philosophically not too problematic, at least as much less problematic as adding e.g. second order quantifiers. We shall show how Tarskian hierarchies may be used for eliminating certain (predicative) second order quantifiers. Hence such hierarchies may form a powerful tool for performing ontological reductions and for eliminating ontological assumptions.

2. NOTATION

It will be convenient to identify expressions and their respective Gödel numbers. We assume that the basic symbols $c, x, f, =, P, T, \neg, \wedge, \exists$, (and) are different numbers, and we use one of the usual primitive recursive encodings $\langle \dots \rangle$ of finite sequences of numbers. The constants are all numbers $\langle c, n \rangle$, the variables the pairs $\langle x, n \rangle$, the k -place function symbols ($k > 0$) the triples $\langle f, k, n \rangle$, where n is arbitrary. The set of all k -place predicate symbols is formed by the triples $\langle P, k, n \rangle$. The set of 2-place predicate symbols includes, in addition, the identity sign $=$, and the set of 1-place predicate symbols all "truth predicates" of the form $\langle T, n \rangle$. The terms are the variables (meta-variables: x, y, z, a, b, v, w), the constants and all sequences $\langle f, t_1, \dots, t_k \rangle$, where f is a k -place function symbol and $t_1, \dots, t_k$ are terms. If P is a k -place predicate symbol, then $\langle P, t_1, \dots, t_k \rangle$ is an atomic formula; if x is a variable and ϕ and ψ are formulas, then $\langle \neg, \phi \rangle$, $\langle (, \phi, \wedge, \psi,) \rangle$ and $\langle \exists, x, \phi \rangle$ are formulas, too⁴. Open and closed (i.e. sentences) formulas are defined in the usual way. When actually writing down formulas we shall obey the usual conventions and write e.g. $\phi \wedge \psi$ instead of $\langle (, \phi, \wedge, \psi,) \rangle$, and for the predicate symbols $\langle T, n \rangle$ we shall use subscripts, which gives T_n .

The constants, function and predicate symbols, except $=$, are the non-logical symbols; all other are the logical ones. A language $\mathcal{L}$ is then determined by its non-logical symbols. Note that the language with logical symbols only is not empty, but consists of those formulas built up with the identity symbol. As we identify a language and its set of formulas, a language is a subset of the set of natural numbers and we can write $\phi \in \mathcal{L}$, if ϕ is a formula of $\mathcal{L}$. A language $\mathcal{L}$ is recursive, if $\mathcal{L}$ is recursive as a set. The following three conditions are obviously equivalent:

- $\mathcal{L}$ is a recursive language.
- The set of non-logical symbols of $\mathcal{L}$ is recursive.
- The properties of being a constant, a function and a predicate symbol of $\mathcal{L}$ are recursive, respectively.

By the language $\mathcal{L}$ expanded by the non-logical symbols $s_1, \dots, s_n$ we understand the language having as non-logical symbols $s_1, \dots, s_n$ and all non-logical symbols of $\mathcal{L}$. In the following we will often encounter the situation where a recursive language $\mathcal{L}$ is expanded by a set of truth predicates T_k , where $k \in I \subseteq \mathbb{N}$. Obviously, the expanded language is recursive, if and only if I is.

The identification is not only a matter of convenience, because we are interested in the property of being a formula of a certain kind to be (intuitively) decidable; and decidability and related notions were successfully explicated in recursion theory, which deals with natural numbers etc., but not with pure uncoded expressions. Of course, we could keep the distinction between formulas and their codes, and ask whether a special set of Gödel numbers of certain kind of formulas is decidable. We shall mention the problems of arithmetization later in this paper again and see that they are by no means trivial; especially, the indexing of truth predicates will be sensitive to the problems of arithmetization. I shall use Church's thesis throughout the paper.

We shall need the special language $\mathcal{L}_{\text{PRA}}$ of the theory of Primitive Recursive Arithmetic PRA having infinitely many function symbols and the constant 0 as non-logical symbols. The indices (natural numbers) of the primitive recursive functions are defined in the usual way. If e is an index, the associated function is designated by $[e]$. Now the function symbols are exactly the triples $\langle f, k, e \rangle$ for all indices e of k -place primitive recursive functions. The symbol $\langle f, k, e \rangle$ will be called a function symbol for the function $[e]$.

If $\mathfrak{M}$ is a model, ϕ a formula and $\mathfrak{b}$ an assignment, that is a mapping from the set of variables into the domain of $\mathfrak{M}$, we shall write $\mathfrak{M} \models \phi$ for “the formula ϕ is valid under the assignment $\mathfrak{b}$ in the model $\mathfrak{M}$ ”. $\mathfrak{M} \models \phi$ (without any assignment) means that ϕ is a sentence and is true in $\mathfrak{M}$.

3. TARSKIAN HIERARCHIES

If we try to build up transfinite Tarskian hierarchies, we choose a language without any truth predicate and expand it first by a new truth predicate T_0 , then by T_1 and so on for all finite levels of the hierarchy. At level ω we should add T_ω , then $T_{\omega+1}$ and so on for all ordinals; but we have to explain what T_ω is going to be. By our conventions the predicate symbol T_ω is a natural number. Hence it cannot be $\langle T, \omega \rangle$ or something embracing the ordinal ω directly. Hence we shall not use ordinals but notations (which are natural numbers) of ordinals instead. In consequence, the number of all truth predicates must be denumerable.

It may be argued that this limitation to countable ordinals is a good reason to dispense with our identification of expressions and natural numbers and to use pairs like $\langle T, \omega \rangle$ as truth predicate where $\langle \dots \rangle$ is now a set theoretic pairing function which is applicable to arbitrary sets including ordinals like ω . If we admitted uncountably many truth predicates, we had to deal with languages whose syntax is not decidable. Furthermore, questions about the complexity (in the sense of recursion theory) of sets of formulas would not make sense any longer. In addition, such languages cannot be considered as really usable languages. But such an approach might be, nevertheless, quite interesting and promising, if it were carried out employing a set-theoretical language as base language to which the truth predicates are added. Because such a construction differs in a very fundamental way from our hierarchies, we shall restrict ourselves to truth predicates which have notations of ordinals as indices. In order to get a notation system for ordinals we need certain orderings of natural numbers.

DEFINITION 1. An index system is a well founded partial ordering, that is a well founded, reflexive, transitive and antisymmetric relation, $\preceq$ of a subset of the natural numbers, such that for all k the set $\{n : n \preceq k\}$ is linearly ordered by $\preceq$.

Obviously, an index system is a well founded tree built up from natural numbers. Note that even the empty set is an index system. As usual, I shall write $i \prec k$ for $i \preceq k \wedge i \neq k$. We are now ready to define the fundamental notion of the paper:

DEFINITION 2. Let $\mathcal{L}$ be a language without a predicate symbol T_n and let $\preceq$ be an index system. Then the Tarski hierarchy over $\mathcal{L}$ with $\preceq$ is the mapping $L : \text{Fld}(\preceq) \rightarrow \text{Pot}(\mathbb{N})$ where the value of $n \in \text{Fld}(\preceq)$ is the language $\mathcal{L}$ expanded by all predicate symbols T_k satisfying $k \prec n$.

In this definition $\text{Pot}(\mathbb{N})$ is the power set of the set of natural numbers and $\text{Fld}(\preceq)$ is the field of the index system, that is the set $\{x : \exists y(x \preceq y \vee y \prec x)\}$, which is by definition identical to the domain $\text{dom}(L)$ of L .

Because of technical reasons I did not exclude ramifications of the index system; but our emphasis is on linear index system. The former do not impose much additional difficulties over the latter. In contrast, index systems which are not well founded have to be treated in a completely different way, because there do not exist any reasonable semantics for the corresponding hierarchies as was shown in (Visser, 1989, p. 637f). Hence we shall neglect them in this paper, although they may have some features making them interesting for the theory of paradoxes; for this see (McCarthy, 1988).

If $\text{Fld}(\preceq)$ is not empty, it contains at least one $\preceq$ -minimal element k and we have $L(k) = \mathcal{L}$ for the corresponding Tarski hierarchy. If L is a Tarski hierarchy, its index system $\preceq_L$ is given by the following equivalence:

$$i \preceq_L k, \text{ if and only if } i, k \in \text{dom}(L) \text{ and } L(i) \subseteq L(k)$$

The following abbreviations are convenient:

$$L(\top) := \bigcup_{n \in \text{dom}(L)} L(n)$$

$$L(\perp) := \begin{cases} \emptyset & \text{if } \text{dom}(L) = \emptyset \\ \bigcap_{n \in \text{dom}(L)} L(n) & \text{else} \end{cases}$$

If L is a Tarski hierarchy over the language $\mathcal{L}$, $L(\perp)$ is the language $\mathcal{L}$. If a number n has no successor in the index system, then T_n is no symbol of the language $L(\top)$; but if it has, it belongs to the language. Hence $L(\top)$ contains $T_n x$, if and only if n is not maximal. Every n in the index system may now be assigned a height in the usual way:

$$|n|_L := \begin{cases} 0 & \text{if } \neg \exists k \, k \prec n \\ \sup\{|k|_L + 1 : k \prec n\} & \text{if } n \text{ is not minimal} \end{cases}$$

The height of the hierarchy is then defined in the obvious way:

$$|L| := |\preceq|_L = \sup\{|n|_L : n \in \text{dom}(L)\}$$

For every countable ordinal and language $\mathcal{L}$ without truth predicate there is trivially a Tarski hierarchy over $\mathcal{L}$ of height α . When we build up a hierarchy over a recursive language with an arbitrary index system, the languages of the hierarchy are not necessarily recursive. In order to get

really usable languages and hierarchies we must impose at least one further condition on the index system. If we are given two truth predicates T_i and T_j , we should be able to tell whether T_i is an “earlier” truth predicate than T_j and whether they are truth predicates of the hierarchy at all. Otherwise the languages of the hierarchy would become unmanageable in the sense that we had to use an oracle in order to decide whether a formula is well formed. These requirements lead to recursive index systems. Note that, if L is a hierarchy of height α with recursive index system and $\text{Fld}(\prec) \neq \mathbb{N}$, there is a hierarchy of height $\alpha + 1$ extending L . So there is no natural halting point for the process of adding a truth predicate and we should not study single hierarchies with recursive index systems but sequences of such hierarchies. We may facilitate this task by considering the Tarski hierarchies which are unions of sequences of hierarchies with recursive index systems. These unions will be called boundedly recursive Tarski hierarchies. Before formally introducing them we shall prove a lemma. For this purpose put for arbitrary $\phi \in L(\top)$:

$$|\phi|_L := \min\{|\phi|_L : \phi \in L(n)\}$$

Hence $\phi \in L(\perp)$ is equivalent to $|\phi|_L = 0$.

LEMMA 3. *Let L be a Tarski hierarchy. Then the following two conditions on L are equivalent:*

- (i) *If $n \in \text{dom}(L)$, then for any two formulas ϕ and ψ of the language $L(n)$ it is decidable whether $|\phi| \leq |\psi|$ holds.*
- (ii) *For every $n \in \text{dom}(L)$ the index system $\preceq$ restricted to the set $\{k : k \prec n\}$ of predecessors of n is recursive.*

The sets $\{k : k \prec n\}$ for $k \in \text{Fld}(\preceq)$ will sometimes be called initial segments of the index system $\preceq$. Obviously, all languages of a hierarchy satisfying one of the above conditions are recursive and (i) may be considered as a uniform version of the requirement that all languages of the hierarchy are recursive.

DEFINITION 4. A Tarski hierarchy is boundedly recursive, if and only if $L(\perp)$ is a recursive language or empty⁵ and (ii) of the preceding lemma is satisfied.

If L is a boundedly recursive hierarchy and n is in $\text{dom}(L)$ then the hierarchy obtained from L by restricting the domain to the set of all $k \preceq n$ is a linear hierarchy with recursive index system. It is also obvious that the union of a possibly infinite sequence $L_0 \subseteq L_1 \subseteq \dots \subseteq L_\alpha$ of hierarchies with recursive index system forms a linear boundedly recursive Tarski hierarchy.

The index system of a boundedly recursive Tarski hierarchy may be arbitrarily complex as well as the field of it. As $n \in \text{Fld}(\preceq)$ holds, if and only if there is a k such that $k \preceq n$ or $n \preceq k$ holds, we are able to prove the following proposition:

PROPOSITION 5. *Assume L is a boundedly recursive Tarski hierarchy. Then the complexity of the domain of L depends in the following way on the complexity of the index system:*

- (i) *If $\preceq$ is Π_n^0 , then $\text{Fld}(\preceq)$ is Σ_{n+1}^0 .*
- (ii) *If $\preceq$ is Δ_n^0 or Σ_n^0 , then $\text{Fld}(\preceq)$ is Σ_n^0 .*
- (iii) *If $\preceq$ is Δ_n^1 , Σ_n^1 or Π_n^1 , then $\text{Fld}(\preceq)$ is of the same complexity.*

The height of a boundedly recursive Tarski hierarchy is limited to ω_1^{CK} , that is the first non-recursive ordinal or the supremum of all order-types of all recursive well-orderings. We are even able to obtain the much stronger result (i) of the following theorem:

THEOREM 6. *If $\mathcal{L}$ is a recursive language without a truth predicate the following holds:*

- (i) *There is a Tarski hierarchy over $\mathcal{L}$ of height α whose index system is Σ_1^1 , if and only if $\alpha \leq \omega_1^{CK}$.*
- (ii) *There is a boundedly recursive Tarski hierarchy over $\mathcal{L}$ of height α , if and only if $\alpha \leq \omega_1^{CK}$.*

Proof. If the index system $\preceq$ is Σ_1^1 , the set of all predecessors $\{k : k \prec n\}$ of a member n of the field of the index system is a linear well-ordering of order-type $|n|_L$. Consequently $|n|_L \geq \omega_1^{CK}$ implies that there is a Σ_1^1 -well-ordering of order-type ω_1^{CK} , which is impossible (see (Rogers, 1967)). So the left-to-right direction of (i) and (ii) is proved.

In order to show the converse direction of (ii) we need a well founded tree of height ω_1^{CK} such that $\preceq$ restricted to $\{k : k \prec n\}$ is recursive for $n \in \text{Fld}(\preceq)$. In Kleene's notation system⁶ $\mathcal{O}$ these initial segments are recursively enumerable, but Girard has provided in (Girard, 1987) a version of $\mathcal{O}$ where its segments are even recursive. This gives the right-to-left direction of both statements.

In boundedly recursive hierarchies it is not in general decidable for any two formulas which of both is lower in the hierarchy, because we may fail to have a decision procedure for telling whether the index of a truth predicate is higher than another given index. In (McGee, 1991, p. 122) it is even proposed not to admit sentences $T_i \overline{T_j} t$ as well formed formulas, if $i \not\preceq j$. If McGee's proposal is accepted and we require all languages of the hierarchy L to be recursive languages, we are forced to consider boundedly

recursive hierarchies only. But as we do not share McGee's view because we do not see a good reason for his restriction, the bounded recursiveness is an additional requirement to the recursiveness of syntax. Of course, such sentences will be evaluated as false in the semantics we shall describe in the next section, but they are perfectly well formed.

One of our main reasons for studying Tarskian hierarchies is the aim to provide a formal explication for a process leading to such hierarchies of object- and metalanguages. We start with a recursive language, construct a metalanguage for it by adding a truth predicate and iterate this process, until we reflect on this process and combine all languages of finite levels to the language of the first limit level ω ; then we carry on this process into the transfinite. When formalizing it we should care that every level of the hierarchy and its languages can really be effectively constructed, which is possible for boundedly recursive hierarchies only. If we try to give axiom systems for the languages of a boundedly recursive hierarchy, it makes good sense to impose even further restrictions⁷ on the index system; but this is beyond the scope of this paper, so we refer to (Feferman, 1991) and (Halbach, 1994).

The above theorem shows the impossibility of carrying on the process of constructing stronger and stronger metalanguages by adding a truth predicate to arbitrarily large countable ordinals. At level ω_1^{CK} it is not sufficient just to add a further predicate, because if we would push the process any further we had to give up decidability of our syntax etc. Nevertheless, it is perfectly possible to use a metalanguage of a hierarchy $L(\omega_1^{CK})$, because otherwise we could not speak about such hierarchies; but it will be of a completely different nature as it cannot contain all languages of the hierarchy. This observation reveals the limitations and shortcomings of our approach: sometimes a metalanguage cannot have all objectlanguages as sublanguages, if we want to keep a recursive metalanguage.

As we mentioned above, we are especially interested into hierarchies whose index system is linear. If we dispense with non-linear hierarchies, we are still able to obtain hierarchies of height ω_1^{CK} ; hence Theorem 6 holds for linear hierarchies, too.

THEOREM 7. *For every recursive language $\mathcal{L}$ without truth predicate there is a boundedly recursive hierarchies over $\mathcal{L}$ whose index system is linearly ordered.*

Proof. To prove the theorem we must exhibit a well-ordering of natural numbers of order-type ω_1^{CK} with recursive initial segments only. Kleene's $\mathcal{O}$ has paths through it with recursively enumerable initial segments. We could use again Girard's version of $\mathcal{O}$, but we can also pick a suitable path

through Kleene's original $\mathcal{O}$. The existence of such a path was shown in (Jokusch, 1975).

4. SEMANTICS

In this section we shall show how to define the semantics of the hierarchies in a formally rigorous way. We shall do this by extending an acceptable structure of $L(\perp)$ to models of the respective languages of the hierarchy. The most important feature of an acceptable model⁸ is the availability of a coding scheme $\langle \dots \rangle^C$ for arbitrary finite sequences of elements of the domain of the structure. The codes $\langle k_1, \dots, k_n \rangle$ of sequences of natural numbers $k_1, \dots, k_n$ should not be confused with the codes $\langle a_1, \dots, a_n \rangle^C$ of arbitrary members of the domain (which always contains by definition all natural numbers or an isomorphic copy), but for natural numbers it is convenient not to distinguish $\langle k_1, \dots, k_n \rangle$ and $\langle k_1, \dots, k_n \rangle^C$, if the arguments $k_1, \dots, k_n$ are natural numbers.

If $\mathfrak{A}$ is a model for a language containing T_n , then $\mathfrak{A}(T_n)$ is the extension of T_n in this model, i.e. the subset of the domain of $\mathfrak{A}$ to which T_n applies. By recursion on the index we define now the models of the hierarchies in the obvious way.

DEFINITION 8. Let L be a Tarski hierarchy and $\mathfrak{A}$ an acceptable structure for $L(\perp)$. The following conditions determine a unique model $\mathfrak{A}_n$ for every language $L(n)$ ($n \in \text{dom}(L)$) of the hierarchy.

- (i) If n is minimal in the index system, $\mathfrak{A}_n$ and $\mathfrak{A}$ are the same model.
- (ii) If n is a direct $\leq_L$ -successor of i , then $\mathfrak{A}_n$ is the unique expansion of $\mathfrak{A}_i$ to $L(n)$ satisfying:

$$\mathfrak{A}_n(T_i) := \{\phi \in L(i) : \mathfrak{A}_i \models \phi\}$$

- (iii) If $|n|_L$ is a limit, we put:

$$\mathfrak{A}_n := \bigcup_{i \prec n} \mathfrak{A}_i$$

Here $\bigcup$ designates the usual union of models (not of arbitrary sets).

Clearly, if $i \prec n$ and k does not belong to the sentences of $L(i)$, $\mathfrak{A}_n(T_i)$ does not contain k .

Sometimes it will be convenient to have a model $\mathfrak{A}_\top$ for the union $L(\top)$ of all languages of a hierarchy. It may be obtained as the union (in the above sense) of all models $\mathfrak{A}_n$ where $n \in \text{dom}(L)$. We are now able to prove a theorem which is analogous to our observation in the first section that Definition (i) of a truth predicate implies Definition (ii).

LEMMA 9 (Tarski biconditionals). *If L is a Tarski hierarchy, $\mathfrak{A}$ an acceptable model of $L(\perp)$, $n \in \text{dom}(L)$ and $\bar{\phi}$ is a name in $\mathfrak{A}$ for $\phi \in L(n)$, then the respective Tarski biconditionals hold, or, formally speaking:*

$$\mathfrak{A}_\top \models T_n \bar{\phi} \leftrightarrow \phi$$

We shall now tackle the question about the complexity of the set of sentences true in $\mathfrak{A}_\top$, that is of the set

$$V := \{\phi : \phi \text{ is a sentence of } L(\top) \text{ and } \mathfrak{A}_\top \models \phi\}.$$

The complexity of V depends crucially on the index system and the base model $\mathfrak{A}$, because the field of the index system and the set $\{\phi \in L(\perp) : \mathfrak{A} \models \phi\}$ are easily seen to be recursive in V . The latter set is hyper elementary according to (Moschovakis, 1974), but not elementary, if $L(\perp)$ contains only finitely many non-logical symbols. This gives a lower bound of the complexity of V for non-empty hierarchies. On the other hand, general results about well founded trees yield the following result.

PROPOSITION 10. *If $|L| \geq \omega_1^{CK}$, then V is at least Π_1^1 .*

Proof. By well-known theorems (see e.g. (Rogers, 1967)), an index system of height ω_1^{CK} or greater is at least Π_1^1 . Let $\prec_L^*$ be the relation $\preceq_L$ restricted to all members of $\text{Fld}(\preceq_L)$ that are not maximal (if the index system has not any maximal elements both relations are the same). Then $\prec_L^*$ is again at least Π_1^1 and can be expressed in the following way:

$$\mathfrak{A}_\top \models \forall x(T_k x \rightarrow T_m x) \iff k \prec_L^* m$$

Because the property of being a sentence of the form $\forall x(T_k \rightarrow T_m)$ is recursive, the set V is at least Π_1^1 , too.

In this way it is always possible to prove that V has at least the complexity of the index system. But such results are more or less trivial, because they rely on theorems about well founded trees and not on the special properties of the truth predicates. So we shall consider hierarchies with sufficiently simple index systems. If the index system is recursive the following theorem may be proved.

THEOREM 11. *Suppose that $\mathfrak{A}$ is an acceptable model for the language $L(\perp)$ containing only finitely many non-logical symbols and that the index system is recursive. Then V is hyper elementary in $\mathfrak{A}$.*

We defer the proof to the next section (see Theorem 13).

The theorem is also valid for languages having infinitely many non-logical symbols, if the property of being an atomic $\mathfrak{A}$ -valid sentence is elementary in $\mathfrak{A}$. For example, this is the case with the language of Primitive Recursive Arithmetic and its standard model. The premises of the theorem concerning the index system can be weakened, too. It will be seen from the proof of the theorem that it is sufficient for the index system to be elementary in $\mathfrak{A}$.

5. DEFINABILITY

If a truth predicate is added to an interpreted language, its expressive power is increased, because by Tarski's Theorem new sets are definable within the expanded language. We shall examine which sets become definable by an iterated expansion using truth predicates. In this section we presuppose that L is a boundedly recursive Tarski hierarchy, $\mathfrak{A}$ an acceptable structure for $L(\perp)$ with domain A .

Remember that all natural numbers are contained in the domain A of the acceptable structure $\mathfrak{A}$. Therefore, if $\mathcal{L}$ is a language for the model $\mathfrak{A}$, we can write $\text{Def}(\mathcal{L})$ for the set of natural numbers definable in $\mathcal{L}$ interpreted by the model $\mathfrak{A}$. As the models for the languages are determined uniquely by $\mathfrak{A}$, the $L(n)$ -definable sets of natural numbers will be designated by $\text{Def}(L(n))$. This means for $n \in \text{dom}(L) \cup \{\top, \perp\}$:

$$\begin{aligned} B \in \text{Def}(L(n)) &\iff B \text{ is definable in } L(n) \\ &\iff \exists \phi(x) \in L(n) \exists b \forall a \in A (a \in B \\ &\iff b(x) = a \wedge \mathfrak{A}_\top \models \phi(x)[b]) \end{aligned}$$

Later in this section we shall use $\text{Def}(\mathcal{L})$ also for other languages $\mathcal{L}$. It will be clear from the context to what model for $\mathcal{L}$ we refer.

Tarski's Theorem can be reformulated using this notation in the following way:

LEMMA 12 (Hierarchy Lemma). $n \prec m$ implies

$$\text{Def}(L(n)) \subsetneq \text{Def}(L(m)).$$

Before analysing what sets are definable we have to introduce some more notation. Within the base language L we need the following predicate and function expressions:

$\text{Var}(x)$	says	" x is a variable"
$\text{At}(x)$	says	" x is an atomic sentence of $\mathcal{L}_{\text{PRA}}$ "
$\text{Ver}(x)$	says	" x is a true atomic sentence of $\mathcal{L}_{\text{PRA}}$ "
$\text{ct}(x)$	says	" x is a closed term of $\mathcal{L}_{\text{PRA}}$ "
$\text{Sent}_{\mathcal{L}}(x)$	says	" x is a sentence of $\mathcal{L}$ "
$\text{val}(x)$	says	"the value of the closed term x "
$\neg x$	says	"the negation of x "
$(x \wedge y)$	says	"the conjunction of x and y "
$\exists vx$	says	"the existential quantification x with the variable v "
$\text{T}_n(x)$	says	"the result of prefixing T_n to x "
$\text{sub}(x, y, z)$	says	"the result of substituting x for y in z "
$\text{num}(x)$	says	"the numeral of x "

It does not matter how the function symbols behave on unsuitable objects, e.g. how $\neg$ behaves on an object which is not a formula. num , which will be needed for arithmetical languages only, applies only to natural numbers. Note that all these properties and functions can be expressed in $L(\perp)$ because the model $\mathfrak{A}$ is acceptable.

By a well-known result the set of true sentences of an acceptable structure $\mathfrak{A}$ is hyperelementary in $\mathfrak{A}$ (see (Moschovakis, 1974) p.67ff). The theorem below shows that the same is true of iterations of truth predicates.

THEOREM 13. *If $L(\perp)$ has only finitely many non-logical symbols,*

- (i) *the extensions $\mathfrak{A}_{\top}(\text{T}_n)$ of the truth predicates T_n are hyperelementary in $\mathfrak{A}$ for non-maximal $n \in \text{dom}(L)$;*
- (ii) *the set of sentences of $L(n)$ valid in $\mathfrak{A}_{\top}$, that is the set*

$$\{\phi \in L(n) : \mathfrak{A} \models \phi\}$$

is hyperelementary for $n \in \text{dom}(L)$.

Theorem 11 is now an obvious consequence of (ii).

Proof. For the proof it is convenient to assume that $L(\perp)$ does not contain any function symbols; they may be thought of as replaced by finitely many suitable new predicate symbols in the usual way. The proof relies on the availability of an elementary coding scheme $\langle \dots \rangle^c$. The languages $\overline{L(n)}$, where $n \in \text{dom}(L)$, are generated, respectively, by adding a constant $\bar{a}$ to $L(n)$ for every $a \in A$. We use for this purpose the pairs $\langle c, a \rangle^c$ consisting of a fixed number c (different from f, P etc.) and the

element a itself. These constants (and all expressions of $\overline{L(n)}$) are again members of the domain A . But note that $\overline{L(n)}$ is no longer a set of natural numbers, because a formula containing such a new constant can fail to be a natural number. So $\overline{L(n)}$ will not be a language in our strict sense. The function assigning to an object x its constant $\langle c, x \rangle^c$ is represented by the function expression c_x .

Apart from the new one-place predicate symbol S the following formula belongs to the language $\overline{L(\perp)}$ for $n \in \text{dom}(n)$.

- $$x \in \overline{L(n)} \wedge$$
- (1) $[\text{At}(x) \wedge \text{Ver}(x) \vee$
 - (2) $\exists y (x = \neg y \wedge \text{At}(y) \wedge \neg \text{Ver}(y)) \vee$
 - (3) $\exists y \exists m \prec \bar{n} (x = \text{T}_m y \wedge \text{Sent}_{\overline{L(m)}}(y) \wedge S y) \vee$
 - (4) $\exists y \exists m \prec \bar{n} (x = \neg \text{T}_m y \wedge \text{Sent}_{\overline{L(m)}}(y) \wedge S \neg y) \vee$
- $$\exists y \exists z (x = (y \wedge z) \wedge S y \wedge S z) \vee$$
- $$\exists y \exists z (x = \neg (y \wedge z) \wedge (S \neg y \vee S \neg z)) \vee$$
- $$\exists v \exists z (x = \exists v z \wedge \exists y \text{Sub}(c_x, v, z)) \vee$$
- $$\exists v \exists z (x = \neg \exists v z \wedge \forall y \text{Sub}(c_x, v, z))]$$

Because $\mathfrak{A}$ is acceptable, we are able to express within $\overline{L(\perp)}$ the function assigning to two formulas their conjunction or the function assigning to a formula its negation etc. Therefore it is possible to quantify over formulas in the above way. Truth for atomic and negated atomic sentences may be defined within $\overline{L(\perp)}$, as $\overline{L(\perp)}$ contains only finitely many predicate symbols (see (Moschovakis, 1974, p. 67f)), hence (1) and (2) are expressible in $\overline{L(\perp)}$. (3) and (4) rely on the bounded recursiveness of the index system, because otherwise the quantifier $\exists m \prec \bar{n}$ could not be used.

Because S occurs only positively in $\phi_n(x, S)$, there is a smallest fixed point $\mathcal{T}_n$ for any $\phi_n(x, S)$. Clearly, $\mathcal{T}_n$ is the set of true sentences of $\overline{L(n)}$. Its complement $\overline{\mathcal{T}_n}$ is inductive, too, as may be seen from the following formula $\overline{\phi}_n(x, S)$:

$$\overline{\phi}_n(x, S) := [x \notin \overline{L(n)} \vee \phi_n((\neg x), S)]$$

As $\mathfrak{A}$ is acceptable the property $x \notin \overline{L(n)}$ is elementary and, in consequence, the set $\mathcal{T}_n$ is coinductive. Because the property $x \in \overline{L(n)}$ is elementary as well, the following set is also coinductive, that is hyperelementary in $\mathfrak{A}$:

$$\{\psi \in L(n) : \mathfrak{A} \models \psi \wedge \psi \text{ is a sentence}\} = \mathcal{T}_n \cap L(n)$$

This proves (ii) of the theorem.

The extensions $\mathfrak{A}_\top(T_n)$ of the truth predicates T_n are exactly the sets of valid sentences of $L(n)$, or in formulas:

$$\{a \in A : \mathfrak{A}_\top \models T_n x[b] \wedge b(x) = a\} = \{\psi \in L(n) : \mathfrak{A}_n \models \psi \text{ and } \psi \text{ is a sentence}\}$$

Hence (i) is established, too.

The above holds also true if “hyper elementary” is replaced by “ Δ_1^1 ” where Δ_1^1 for arbitrary structures is defined as in (Moschovakis, 1974), because all hyper elementary sets are Δ_1^1 . But note that we have admitted also uncountable structures, so both notions do not coincide. This remark also applies to the following corollary on definability.

COROLLARY 14. *If $L(\perp)$ contains only finitely many non-logical symbols, then all sets $B \in \text{Def}(L(n))$ are hyper elementary for every $n \in \text{dom}(L)$.*

Proof. The extensions of the truth predicates are hyper elementary. Hence all $B \in \text{Def}(L(n))$ are elementary in hyper elementary parameters; so they are hyper elementary themselves.

For (3) and (4) in the proof of the theorem it is only required that all initial segments of the index system are elementary. So the theorem can be proved for “boundedly elementary” hierarchies as well, and we even get the following strengthened version of the theorem:

COROLLARY 15. *If the index system is elementary in $\mathfrak{A}$ and $L(\perp)$ contains only finitely many non-logical symbols, then the set of sentences of $L(\top)$ valid in $\mathfrak{A}_\top$ is hyper elementary in $\mathfrak{A}$.*

There is also a second possibility to weaken the premises of the theorem. The assumption that $L(\perp)$ contains only finitely many non-logical symbols was needed only for the definability of the true atomic and negated atomic sentences of $L(\perp)$ in $L(\perp)$. So if the properties of being a true atomic and negated atomic sentences of $L(\perp)$ are elementary the assumption concerning the finiteness of non-logical symbols may be dropped; so the theorem pertains also to the language $\mathcal{L}_{\text{PRA}}$.

From the theorem we know that all sets definable in a language $L(n)$ of the hierarchy ($n \in \text{dom}(L)$) is hyper elementary. But what about the inverse inclusion? Are all hyper elementary sets definable in some language $L(n)$, if the index system and the base model $\mathfrak{A}$ are chosen in a suitable way? The rest of the paper is devoted to a proof of the positive answer to this question.

Of course, the most straightforward choice is $L(\perp) = \mathcal{L}_{\text{PRA}}$ (or some other arithmetical language) together with the standard model. The order-type of the index system must be as high as possible, that is ω_1^{CK} , in order to define all hyperelementary, in this case all hyperarithmetical, sets; Kleene's $\mathcal{O}$ will be employed, because it occurs also in the definition of the ramified analytical hierarchy and it is convenient, if the index system of the Tarski hierarchy and the index system of the ramified analytical hierarchy coincide (this will become clear in the proofs).

We shall show that definability in this hierarchy corresponds⁹ to membership in the ramified analytical hierarchy and therefore in the hyperarithmetical hierarchy. Our result will prove that we can dispense with predicative second order quantifiers, if the truth predicates of the mentioned Tarski hierarchy are available.

Because many different versions of $\mathcal{O}$ may be found in the literature, I shall fix one. For this purpose, let $\{e\}$ the partial recursive function which has (among others) the index e .

DEFINITION 16. $<_{\mathcal{O}}$ is the smallest relation on $\mathbb{N}$ having the following properties:

- (i) $1 <_{\mathcal{O}} 2$.
- (ii) If $c <_{\mathcal{O}} d$, then $c <_{\mathcal{O}} 2^d$.
- (iii) If $\{e\}$ is a total recursive function and $n \in \mathbb{N}$ $\{e\}(n) <_{\mathcal{O}} \{e\}(n+1)$ holds for all numbers n , then $\{e\}(n) <_{\mathcal{O}} 3 \cdot 5^e$ for all n .
- (iv) $<_{\mathcal{O}}$ is transitive.

The field $\{n : \exists d \ n <_{\mathcal{O}} d\}$ of $<_{\mathcal{O}}$ is called $\mathcal{O}$. $<_{\mathcal{O}}$ is easily shown to be a well founded tree and therefore an index system. In accordance with usual practice we shall write $|d|_{\mathcal{O}}$ for $|d|_{<_{\mathcal{O}}}$.

Because a Tarski hierarchy is determined by the base language $L(\perp)$ and its index system, the following definition is sound:

DEFINITION 17. Let $N : \mathcal{O} \rightarrow \text{Pot}(\mathbb{N})$ be the hierarchy with following properties:

- (i) $N(0) := \mathcal{L}_{\text{PRA}}$
- (ii) $N(2^d)$ is the language $N(d)$, expanded by the truth predicate T_d .
- (iii) $N(3 \cdot 5^e) := \bigcup \{N(\{e\}(n)) : n \in \mathbb{N}\}$

If $\mathfrak{N}$ is the standard model of $\mathcal{L}_{\text{PRA}}$, the models $\mathfrak{N}_d$ are determined by Definition 8. In order to give a characterization of the sets $\text{Def}(N(d))$ we introduce the ramified analytical hierarchy. We shall have to refer to single variables for its definition and for the proofs to come. Instead of $\langle x, n \rangle$ we shall simply write x_n . The variable x_2 will play a special role. In order to emphasize this we shall write y for x_2 .

DEFINITION 18. For every $d \in \mathcal{O}$ a language $\mathcal{L}_d^{2M}$ is defined by recursion on $<_{\mathcal{O}}$. Simultaneously, I shall define the standard model for the union $\bigcup \{\mathcal{L}_d^{2M} : d \in \mathcal{O}\}$ of these languages. For simplicity, we write $\mathfrak{N} \models \phi[b]$ to indicate that ϕ is valid in $\mathfrak{N}$ under the assignment b .

- (i) $d = 1$. $\mathcal{L}_1^{2M}$ is the language $\mathcal{L}_{\text{PRA}}$.
- (ii) $d = 2^c$. $\mathcal{L}_c^{2M}$ is a sublanguage of $\mathcal{L}_d^{2M}$. $\mathcal{L}_d^{2M}$ contains in addition all second order variables X_n^c for all $n \in \mathbb{N}$. Furthermore, there is a set constant $\overline{\Phi(y)}$ for every formula $\Phi(y) \in \mathcal{L}_c^{2M}$ having exactly y as free variable. The additional atomic formulas of $\mathcal{L}_d^{2M}$ are of the form $t \in X_n^c$ and $t \in \overline{\Phi(y)}$ for all variables X_n^c and constants $\overline{\Phi(y)} \in \mathcal{L}_c^{2M}$. The language $\mathcal{L}_d^{2M}$ is closed under existential quantification with the variables X_n^d and under the usual first order operations. The semantics of the new formulas are given in the following way:

$$\begin{aligned}\mathfrak{N} \models t \in \overline{\Phi(y)}[b] &\iff \mathfrak{N} \models \Phi(t)[b] \\ \mathfrak{N} \models \exists X_n^c \Phi(X_n^c)[b] &\iff \exists \Phi(y) \in \mathcal{L}_c^{2M} \mathfrak{N} \models \Phi(\overline{\Phi(y)})[b]\end{aligned}$$

These two clauses define $\mathfrak{N} \models \Phi[b]$ for all formulas $\Phi \in \mathcal{L}_d^{2M}$, as the rules for the variables and constants of lower level, the first-order quantifier and the connectives are not changed.

- (iii) $d = 3 \cdot 5^e$. In this case put:

$$\mathcal{L}_d^{2M} := \bigcup_{n \in \mathbb{N}} \mathcal{L}_{\{e\}(n)}^{2M} = \bigcup_{c <_{\mathcal{O}} d} \mathcal{L}_c^{2M}$$

Validity for all formulas of this language has been already defined.

The uppercase Greek letters Φ, Ψ etc. range over formulas of the languages $\mathcal{L}_d^{2M}$. Usually these languages are defined without the constants¹⁰ $\overline{\Phi(y)}$. But it is easily seen that these constants do not increase the expressive power of the languages; that is, all sets definable in this languages are definable without constants. The language obtained from $\mathcal{L}_d^{2M}$ by dropping all set constants $\overline{\Phi(y)}$ will be designated by $\mathcal{L}_d^2$. If $\text{Def}(\mathcal{L}_d^2)$ and $\text{Def}(\mathcal{L}_d^{2M})$ are the classes of sets definable in $\mathcal{L}_d^2$ and $\mathcal{L}_d^{2M}$, respectively, and $|d|_{\mathcal{O}} = |b|_{\mathcal{O}}$ we have therefore:

$$\text{Def}(\mathcal{L}_d^2) = \text{Def}(\mathcal{L}_d^{2M}) = \text{Def}(\mathcal{L}_b^{2M})$$

The independence from the notation, i.e. the last equation above follows directly from the definition of these languages. The ramified analytical hierarchy may be considered as a bridge¹¹ between the Tarski hierarchy N and a better known hierarchy in recursion theory, namely the hyperarithmetical hierarchy. In (Kleene, 1959) Kleene showed that the sets in the ramified analytical hierarchy up to ω_1^{CK} are exactly the hyperarithmetical sets. So we get from Corollary 14 the following identity:

COROLLARY 19. $\bigcup_{d \in \mathcal{O}} \text{Def}(N(d)) \subseteq \Delta_1^1 = \bigcup_{d \in \mathcal{O}} \text{Def}(\mathcal{L}_d^2)$

Proof. $\mathcal{L}_{\text{PRA}}$ has infinitely many function symbols. But the sets of all true atomic and of all negated atomic sentences of this language are arithmetical (and even recursive), respectively, and therefore Corollary 14 applies (see the remark to this corollary). Hence all sets definable in N are hyperarithmetical, that is Δ_1^1 .

We now want to prove that all hyperarithmetical sets are definable within N . We shall establish an even stronger result by exhibiting a correspondence between each level of the Tarski hierarchy N and the ramified analytical hierarchy, that is

$$\text{Def}(N(d)) = \text{Def}(\mathcal{L}_d^2).$$

A proof of this will be achieved by a translation of the language $N(d)$ into $\mathcal{L}_d^2$ and vice versa. First we construct a translation from $N(d)$ into $\mathcal{L}_d^{2M}$. In (Kleene, 1944) Kleene constructed a recursively enumerable relation $\prec$ coinciding with $<_{\mathcal{O}}$ for elements of $\mathcal{O}$; i.e., if $d \in \mathcal{O}$, $\forall n (n <_{\mathcal{O}} d \Leftrightarrow n \prec d)$ holds. This relation is represented by $x \prec y$.

The degree of a sentence $\phi \in N(\top)$ is defined in the usual way: atomic sentences of $\mathcal{L}_{\text{PRA}}$ and sentences $T_d t$ are of degree 0, while connectives and the quantifier increase the degree by 1. The function assigning each formula its degree is represented by the function symbol $\text{grad}(x)$.

LEMMA 20 (*d*-Comprehension). *If $d \in \mathcal{O}$, the following scheme¹² d -CA is valid in $\mathfrak{N}$:*

$$\exists X^d \forall x (x \in X^d \leftrightarrow \Phi(x))$$

Here $\Phi(x)$ must be a member of $\mathcal{L}_{2^d}^{2M}$ and in $\Phi(x)$ no bound variables X_n^d are allowed; the variable X^d itself must not occur in Φ at all.

In addition, we need the following rule, which is the semantical counterpart to the “limit generalization rule” in (Feferman, 1964):

LEMMA 21 (limit rule). *Suppose $c \in \mathcal{O}$ and $|c|_{\mathcal{O}}$ is a limit ordinal. Then we have for all $\Phi \in \bigcup \{\mathcal{L}_d^{2M} : d \in \mathcal{O}\}$*

$$\exists c <_{\mathcal{O}} d \quad \mathfrak{N} \models \exists X^c \Phi(X^c) \quad \Longleftrightarrow \quad \mathfrak{N} \models \exists X^d \Phi(X^d).$$

In the next lemma we shall give a truth definition for $N(d)$ within the language $\mathcal{L}_d^2$. We shall use a predicate similar to that in (Takeuti, 1987, p. 183f) and iterate the construction.

LEMMA 22. For all $d \in \mathcal{O}$ there is a formula $\Theta_d(x) \in \mathcal{L}_{2^d}^2$ satisfying for all $n \in \mathbb{N}$:

$$\mathfrak{N}_T \models T_d \bar{n} \iff \mathfrak{N} \models \Theta_d(\bar{n})$$

Proof. Again, we use induction on $<_{\mathcal{O}}$.

$d = 2^c$. (The case $d = 1$ is treated in a similar way, so we skip it.) By induction hypothesis, there is a formula $\Theta_c(x) \in \mathcal{L}_{2^c}^2 = \mathcal{L}_d^2$ for which the following equivalence holds:

$$\mathfrak{N}_T \models T_c \bar{n} \iff \mathfrak{N} \models \Theta_c(\bar{n})$$

By $\Phi_d(X, v)$ the formula below is abbreviated:

$$\begin{aligned} & \Theta_c(v) \vee [\forall a (\text{At}_0(a) \wedge \text{grad}(a) = 0 \rightarrow (a \in X \leftrightarrow \text{Ver}_0(a)) \wedge \\ & \forall a (\neg a \in X \leftrightarrow a \notin X \wedge \text{Sent}_{N(d)}(a) \wedge \text{grad}(\neg a) \leq v) \wedge \\ & \forall a \forall b (a \wedge b \in X \leftrightarrow a \in X \wedge b \in X \wedge \text{grad}(a \wedge b) \leq v) \wedge \\ & \forall a \forall w (\exists w a \in X \leftrightarrow \exists z \text{sub}(\text{num}(z), w, a) \in X \wedge \text{grad}(\exists w a) \\ & \leq v \wedge \text{Var}(w)) \wedge \\ & \forall a \forall x (\text{T}_x a \in X \leftrightarrow \exists z (\text{ct}(a) \wedge \text{val}(a) = z \wedge \Theta_c(z) \wedge \text{Sent}_{N(x)} \\ & (z) \wedge x \prec \bar{d})) \wedge \\ & \forall a (a \in X \rightarrow \text{Sent}_{N(\bar{d})}(x))] \end{aligned}$$

$\Phi_d(X, v)$ says that X is the set of all true sentences of $N(c)$. By induction on the degree n we are able to prove this formally:

$$(5) \quad \text{If } \mathfrak{N} \models \exists X^d (\Phi_d(X^d, \bar{k}) \wedge \bar{n} \in X^d), \text{ then } n \in N(d) \text{ is true.}$$

If $n \in N(d)$ is true and the degree of n is at most k , then the following holds:

$$(6) \quad \mathfrak{N} \models \forall X^d (\Phi_d(X^d, \bar{k}) \rightarrow \bar{n} \in X^d)$$

Therefore (5) and (6) show that the formulas $\Phi_d(X, v)$ are adequate. But it is left to prove that such a set X to which $\Phi_d(X, v)$ applies really exists, that is,

$$(7) \quad \forall k \mathfrak{N} \models \exists X^d \Phi_d(X^d, \bar{k})$$

has to be shown. This will be done by induction on k .

$k = 0$. By d -CA (Lemma 20) we get:

$$\begin{aligned} \mathfrak{N} \models & \exists X^d \forall x [x \in X^d \leftrightarrow \\ & \text{Ver}_0(x) \vee \Theta_c(x) \vee \\ & \exists y \prec \bar{d} \exists z (\text{ct}(x) \wedge x = \text{T}_y z \wedge \exists w (\text{val}(z) = w \wedge \Theta_c(w) \wedge \\ & \text{Sent}_{N(y)}(w))] \end{aligned}$$

and hence $\mathfrak{N} \models \exists X^d \Phi(X^d, \bar{0})$.

$k = i + 1$. By induction hypothesis $\mathfrak{N} \models \exists X^d \Phi_d(X^d, \bar{i})$, i. e. there is $\mathbf{M} \in \text{Def}(\mathcal{L}_d^2)$ satisfying $\mathfrak{N} \models \Phi_d(\bar{\mathbf{M}}, \bar{i})$; so there exists a formula $\Psi(x) \in \mathcal{L}_d^2$ such that

$$\mathfrak{N} \models \Psi(\bar{n}) \iff \mathfrak{N} \models \bar{n} \in \bar{\mathbf{M}}$$

Let $\Pi(x) \in \mathcal{L}_d^2$ be the following formula:

$$\begin{aligned} \Psi(x) \vee \text{grad}(x) = \bar{k} \wedge [& \exists a(x = \neg a \wedge \Psi(a)) \vee \\ & \exists a \exists b(x = (a \wedge b) \wedge \Psi(a) \wedge \Psi(b)) \vee \\ & \exists a \exists v(\text{Var}(v) \wedge x = \exists v a \wedge \\ & \exists w \Psi(\text{sub}(\text{num}(w), v, a)))] \end{aligned}$$

$\Pi(x)$ may be shown to be a truth definition for sentences of complexity k :

$$(8) \quad \mathfrak{N} \models \forall X^d [\Phi_d(X^d, \bar{k}) \leftrightarrow \forall x \in X^d \Pi(x)]$$

By d -CA (Lemma 20) the existence of an appropriate set is established:

$$\mathfrak{N} \models \exists X^d \forall x(x \in X^d \leftrightarrow \Pi(x))$$

From this and (8) we conclude:

$$\mathfrak{N} \models \exists X^d \Phi(X^d, \bar{k})$$

Hence we are done with the proof of (7). $\Theta_d(x)$ is the following formula:

$$\mathfrak{N} \models \exists X^d (\Phi_d(X^d, \text{grad}(x)) \wedge x \in X^d).$$

The claim follows now from (5) and (6), because $\mathfrak{N} \models T_d \bar{n}$ holds, if and only if $n \in N(d)$ is true or, formally speaking, because

$$\mathfrak{N}_\top \models T_d \bar{n}, \text{ if and only if } \mathfrak{N} \models \Theta_d(\bar{n}).$$

We have to consider the case where d is a limit notation. So assume $d = 3 \cdot 5^e$. Let $\Xi(X)$ be the following formula:

$$\begin{aligned} \exists c \prec \bar{d} [& \forall x(\text{grad}(x) = \bar{0} \wedge \text{Sent}_{N(\bar{1})}(x) \rightarrow (x \in X \leftrightarrow \text{Ver}_0(x))) \wedge \\ & \forall x(\neg x \in X \leftrightarrow \text{Sent}_{N(c)}(x) \wedge x \notin X) \wedge \\ & \forall x \forall y((x \wedge y) \in X \leftrightarrow x \in X \wedge y \in X) \wedge \\ & \forall x \forall v(\exists v x \in X \leftrightarrow \text{Var}(v) \wedge \exists z \text{sub}(\text{num}(z), y, x) \in X) \wedge \\ & \forall x \forall z(\text{T}_z x \in X \leftrightarrow z \prec c \wedge \text{ct}(x) \wedge \exists y(\text{val}(x) = y \wedge \\ & \text{Sent}_{N(z)}(y) \wedge y \in X)) \wedge \\ & \forall x(x \in X \rightarrow \text{Sent}_{N(c)}(x))] \end{aligned}$$

Put $M_c := \{n \in N(c) : \mathfrak{N}_\top \models n\}$. If $c <_{\mathcal{O}} d$, $\text{Def}(\mathcal{L}_c^2)$ contains by induction hypothesis this set M_c . If $\Phi(y) \in \mathcal{L}_d^2$ is assumed, we can prove that

$$\mathfrak{N} \models \Xi(\overline{\Phi(y)}), \text{ if and only if } \exists c <_{\mathcal{O}} d \{n \in \mathbb{N} : \mathfrak{N} \models \Phi(n)\} = M_c;$$

from this we get

$$\begin{aligned} \exists c <_{\mathcal{O}} d \mathfrak{N} \models \exists X^c (\Xi(X^c) \wedge \bar{n} \in X^c) &\iff \exists c <_{\mathcal{O}} d (n \in \\ &N(c) \wedge \mathfrak{N}_\top \models n) \\ &\iff n \in N(d) \text{ is true.} \end{aligned}$$

Using the limit rule 21 the left-hand side of this equivalence may be reformulated in the following way:

$$\mathfrak{N} \models \exists X^d (\Xi(X^d) \wedge \bar{n} \in X^d) \iff n \in N(d) \text{ is true}$$

So $\exists X^d (\Xi(X^d) \wedge x \in X^d)$ may be used as $\Theta_d(x)$, and we are done with the case $d = 3 \cdot 5^e$, too.

The lemma implies the following corollary which will be needed in the proof of Theorem 25:

COROLLARY 23. *If $c \in \mathcal{O}$, then $\text{Def}(N(c)) \subseteq \text{Def}(\mathcal{L}_c^2)$.*

Now we shall show how to define membership and second order variables within $N(d)$ in order to establish the identity between $\text{Def}(N(c))$ and $\text{Def}(\mathcal{L}_c^2)$.

There is a primitive recursive function $f : \mathbb{N}^2 \rightarrow \mathbb{N}$ having the following property:

$$f(n, k) = \text{the numeral of } (\phi(t(\dot{v}_1, \dots, \dot{v}_j))),$$

if n is the formula $\phi(y) \in \bigcup \{N(d) : d \in \mathcal{O}\}$ and k the term $t(v_1, \dots, v_j)$. The dots above the variables indicate that they may be bound from “outside” (this may be achieved using the substitution and numeral function.). It is assumed that the variables in $\phi(y)$ are renamed in order to avoid a collision with $(v_1, \dots, v_j)$. Hence $f(n, k)$ has $v_1, \dots, v_j$ as free variables. We need a function symbol $\dot{f}$ representing f in $\mathcal{L}_{\text{PRA}}$ (there are infinitely many such symbols). We are now ready to define the translation function.

DEFINITION 24. $*$ is a function on the sets of the terms and formulas of $\bigcup \{\mathcal{L}_\top^{\in \mathcal{M}} : \top \in \mathcal{O}\}$ into set of the terms and formulas of $N(\top)$. It is given by the following clauses:

- (i) $x_n^* := x_{5^n}$ (the 5^n -th variable)
- (ii) If t is a term or an atomic formula of $\mathcal{L}_{\text{PRA}}$, t^* is the result of substituting all variables x by x^* in the term t .
- (iii) $(X_n^d)^* := x_{3^d \cdot 5^n}$, that is the $3^d \cdot 5^n$ -th variable.
- (iv) $(t \in X_n^d)^* := T_d f((X_n^d)^*, t^*)$
- (v) $(t \in \overline{\Phi(y)})^* := T_{2^d} f(\overline{(\Phi(y))^*}, t^*)$, where d is according to $<_{\mathcal{O}}$ the greatest index occurring as an index of a second order variable X_n^d .
- (vi) $(\neg \Phi)^* := \neg \Phi^*$ und $(\Phi \wedge \Psi)^* := \Phi^* \wedge \Psi^*$
- (vii) $(\exists x \Phi(x))^* := \exists x^* (\Phi(x))^*$, if x is a first-order variable.
- (viii) $(\exists X \Phi(X))^* := \exists X^* (\text{Form}(X^*, c) \wedge (\Phi(X))^*)$, if X is a variable X_n^c . $\text{Form}(x, c)$ is a formula¹³ of $\mathcal{L}_{\text{PRA}}$ saying that x is a formula of $N(c)$ with only y free.
- (ix) If none of the above cases applies to n , set $n^* = n$.

If $\mathfrak{b}$ is an assignment, then the assignment $\mathfrak{b}^*$ is obtained from $\mathfrak{b}$ by modifying $\mathfrak{b}$ in the following way: put $\mathfrak{b}^*(v) = \mathfrak{b}(v)$ for all first order variables v ; for all other variables $\mathfrak{b}^*$ coincides with $\mathfrak{b}$.

Note that $v^* \neq y$ for all first or second order variables v .

THEOREM 25. For $d \in \mathcal{O}$ and all formulas $\Phi \in \mathcal{L}_d^{2M}$

$$\mathfrak{N} \models \Phi[\mathfrak{b}] \iff \mathfrak{N}_{\top} \models \Phi^*[\mathfrak{b}^*].$$

From this we get in particular $\text{Def}(\mathcal{L}_d^{2M}) \subseteq \text{Def}(L(d))$.

Proof. The proof is by induction on $<_{\mathcal{O}}$ and side induction on the complexity of Φ .

$d = 1$. In this case $\Phi \in \mathcal{L}_{\text{PRA}}$ and Φ^* is only an alphabetical variant of Φ , that is Φ^* and Φ are different at most with respect to the variables occurring in them.

$d = 2^c$. If $\Phi \in \mathcal{L}_d^{2M} - \mathcal{L}_c^{2M}$ is a sentence $t \in \overline{\Psi(y)}$, where t is a closed term of $\mathcal{L}_{\text{PRA}}$, we have:

$$\begin{aligned} \mathfrak{N} \models t \in \overline{\Psi(y)}[\mathfrak{b}] &\iff \mathfrak{N} \models \Psi(t)[\mathfrak{b}] && (\text{Def.18}) \\ &\iff \mathfrak{N}_{\top} \models \Psi^*(t^*)[\mathfrak{b}^*] && (\text{I.H.}) \\ &\iff \mathfrak{N}_{\top} \models T_d f(\overline{\Phi^*(y)}, t^*)[\mathfrak{b}^*] && (\text{Lem. 9}) \\ &\iff \mathfrak{N}_{\top} \models \Phi^*[\mathfrak{b}^*] && (\text{Def. 24 (v)}) \end{aligned}$$

So the claim is established for atomic sentences. Next, we consider the case $\Phi = \exists X_n^c \Psi(X_n^c)$.

$$\begin{aligned} \mathfrak{N} \models \exists X_n^c \Psi(X_n^c)[\mathfrak{b}] &\iff \exists \Xi(y) \in \mathcal{L}_c^{2M} \mathfrak{N} \models \Psi(\overline{\Xi(y)})[\mathfrak{b}] && (\text{Def.18}) \\ &\iff \exists \Xi(y) \in \mathcal{L}_c^{2M} \mathfrak{N} \models (\Psi(\overline{\Xi(y)}))^*[\mathfrak{b}^*] && (\text{I.H.}) \\ &\iff \exists \phi(y) \in L(c) \mathfrak{N} \models (\Psi^*(\overline{\phi(y)}))[\mathfrak{b}^*] && (\dagger) \\ &\iff \mathfrak{N} \models \exists z (\text{Form}(z, c) \wedge (\Psi^*(z)))[\mathfrak{b}^*] \\ &\iff \mathfrak{N} \models (\exists X_n^c \Psi(X_n^c))^*[\mathfrak{b}^*] && (\text{Def. of } *) \end{aligned}$$

Because $v^* \neq u^*$ for any two different variables v and u , $z = (X_n^c)^*$ is not bound by a “wrong” quantifier.

In line marked (†) we need to know that $\text{Def}(N(c)) = \text{Def}(\mathcal{L}_c^2)$. The direction $\subseteq$ holds by Corollary 23, the other by induction hypothesis.

We skip the cases $\neg\Psi$ and $\Psi \vee \Theta$, because they are trivial. So we are finished with the successor case.

$d = 3 \cdot 5^e$. If $\Phi \in \mathcal{L}_d^{2M}$, there is a $c <_{\mathcal{O}} d$ satisfying $\Phi \in \mathcal{L}_c^{2M}$ and the claim holds by induction hypothesis.

This lemma proves that all hyperarithmetical sets are definable in the Tarski hierarchy N .

THEOREM 26. $\bigcup_{d \in \mathcal{O}} \text{Def}(N(d)) = \Delta_1^1$

Proof. Consider the following chain of inclusions:

$$\Delta_1^1 = \bigcup_{d \in \mathcal{O}} \text{Def}(\mathcal{L}_d^{2M}) \subseteq \bigcup_{d \in \mathcal{O}} \text{Def}(N(d)) \subseteq \Delta_1^1$$

The first identity is guaranteed by Corollary 19, the first inclusion by the preceding theorem and the last by Theorem 14.

Lemma 25 and Lemma 22 show that the correspondence is level by level:

THEOREM 27. *Suppose $|d|_{\mathcal{O}} = \alpha$. Then the following holds:*

$$\text{Def}(N(d)) = \text{Def}(\mathcal{L}_d^2) = \mathcal{H}_{\omega(1+\alpha)}.$$

Here $\mathcal{H}_\alpha$ is the α -th layer of the hyperarithmetical hierarchy (see (Rogers, 1967) or (Sacks, 1990)). The second identity is an old result due to Kleene (Kleene, 1959). In particular, the theorem implies that definability in the Tarskian hierarchy N does not depend on the notation, but only on the ordinal level:

COROLLARY 28 (extensionality). *If $|c|_{\mathcal{O}} = |d|_{\mathcal{O}}$, then $\text{Def}(N(c)) = \text{Def}(N(d))$.*

The proof of this lemma relies on the independence of the particular notation of the ramified analytical hierarchy shown in (Kleene, 1959) (it follows from the second identity in the theorem). Whereas the extensionality of the ramified analytical hierarchy is more or less obvious, because we could even use ordinals directly, the corresponding result for Tarski hierarchies is not established so easily, because the notation system is used in an essential

way for defining the Tarski hierarchy. By analysing the proof of the theorem we can strengthen the above corollary. It may be easily checked that we could have used any other index system, if there were an arithmetical relation coinciding with the index system on its field. Kleene's relation $\prec$ was used in the case of $\prec_{\mathcal{O}}$. We can even go further: we have used Kleene recursively enumerable relation $\prec$ only in contexts like $\dots \prec \bar{d}$ where $\bar{d}$ is a fixed numeral. So we can prove the theorem for any index system which has only arithmetical initial segments; in particular, it holds for any boundedly recursive Tarski hierarchy L with $L(\perp) = \mathcal{L}_{\text{PRA}}$ and the standard model. In such a hierarchy all hyperarithmetical sets are definable, if and only if it has height ω_1^{CK} .

Further, much less trivial generalizations concern hierarchies exceeding ω_1^{CK} . By Theorem 6 some of the initial segments of such index systems will be of complexity Π_1^1 at least, and the proof of the above theorem does not apply to such "large" Tarskian hierarchies. In contrast, it is quite easy to extend the ramified analytical hierarchy further, until its fixed point β_0 is obtained (this was done by Putnam and others, see (Boyd *et al.*, 1969)), because the indexing of the second order variables is quite inessential. But we conjecture that a match-up could be obtained for higher ordinal levels of the Tarskian and the ramified analytical hierarchy.

In (Moschovakis, 1974, p. 120ff) it has been shown how to build up the ramified analytical hierarchy (ramified second order hierarchy) over an arbitrary acceptable structure. It may be shown to be equivalent to the hierarchy generated by the corresponding Tarski hierarchy. But in order to be able to define all hyperarithmetical sets the Tarski hierarchy has to be extended to the ordinal of the respective structure. So we are again faced with the problem of how to extend the Tarski hierarchy smoothly beyond ω_1^{CK} .

Theorem 27 may also be considered as a result on ontological reduction: All hyperarithmetical sets may be defined within the language of arithmetic augmented by suitably many truth predicates; hence we can dispense with the quantifiers, which are usually used to define them. For some technical and philosophical reasons we would prefer reduction of axiomatic theories instead of interpreted languages. Although no precise account of the axioms systems for the Tarskian hierarchy was given in (Feferman, 1991), the equivalence of these systems and the corresponding theories for the ramified analytical hierarchy was stated, these are systems of ramified analysis. We hope that our semantical considerations are detailed enough to render the proof of this fact up to ω routine. For limit levels it is still unclear how a natural axiom of the Tarskian hierarchy should look like.

Extensive results have been proved on definability in certain models of type-free truth. Especially (Burgess, 1986) and (McGee, 1991) contain some advanced results. The latter also gives some information on hierarchies which are somewhat different from the Tarskian hierarchies studied in this paper. Our results, in particular Theorem 27, invite now a further comparison with these type-free conceptions of truth. A detailed comparison of the minimal fixed point model of (Kripke, 1975) based on the strong Kleene evaluation scheme and our hierarchies may be found in (Halbach, 1997).

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NOTES

¹ Tarski clearly favoured the syntactical approach and used the term “language” for what we would now call a theory. See also (Visser, 1989, p. 633).

² It is well-known that this condition may be weakened by requiring only that $\mathcal{L}_{\mathcal{M}}$ contains translations of all sentences of $\mathcal{L}_{\mathcal{O}}$. The definition of a metalanguage has to be modified according to this more general approach.

³ A similar fact was already mentioned by Tarski. He noticed (Tarski, 1935, p. 381) that the Law of Contradiction $\neg\tau(\phi) \vee \neg\tau(\neg\phi)$ (or “ $x \in W_r$ oder $\overline{x} \in W_r$ ” in Tarski’s terms) is derivable from the Tarski biconditionals for each single sentence ϕ , while the corresponding universal sentence is not.

⁴ Other connectives and the universal quantifier may be defined as usual.

⁵ If we did not admit $L(\perp) = \emptyset$ in the definition the empty function would not be a recursively bounded hierarchy and this would cause some inconvenience when stating some of theorems about such hierarchies.

⁶ We shall formally introduce this notation system in Definition 16. For proofs of results on $\mathcal{O}$ we shall need see (Rogers, 1967) or (Sacks, 1990).

⁷ An important and philosophically interesting restriction is autonomy; for arithmetical systems this will bind the order-type of index systems to Γ_0 if reasonable systems are employed.

⁸ For a definition of an acceptable structure see (Moschovakis, 1974, p. 65f).

⁹ Feferman stated implicitly in his paper (Feferman, 1991) a similar result without proof.

¹⁰ Because of these constants $\overline{\Phi(y)}$ it is necessary to define the languages and validity simultaneously. See the original paper (Kleene, 1959) or, for a more modern approach, (Sacks, 1990).

¹¹ In (Church, 1976) a kind of equivalence between ramified type theory and Tarskian hierarchies was established.

¹² These comprehension principles are used in the axiom systems for ramified analysis. Hence I shall call them *d*-CA (*d*-comprehension axiom), too.

¹³ Kleene's recursively enumerable relation $\prec$ is needed to define such a formula $\text{Form}(x, c) \in \mathcal{L}_{\text{PRA}}$, and it may be assumed to be a Σ_1^0 -formula. For other index systems possessing a recursive relation corresponding to $\prec$ the translation function $*$ may shown to be recursive by using some form of the Recursion Theorem. This observation is useful, if we apply $*$ to get proof-theoretical reductions.

REFERENCES

- Boyd R., Hensel, G. and Putnam, H.: 1969, 'A Recursion-Theoretic Characterization of the Ramified Analytical Hierarchy', *Transactions of the American Mathematical Society* **141/142**, 37–62.
- Burge, T.: 1979, 'Semantical Paradox', *Journal of Philosophy* **76**, 83–118, reprinted in (Martin, 1984).
- Burgess, J.: 1986, 'The Truth is Never Simple', *Journal of Symbolic Logic* **51**, 663–681.
- Church, A.: 1976, 'Comparison of Russell's Resolution of the Semantical Antinomies with that of Tarski', *Journal of Symbolic Logic* **41**, 747–760, reprinted in (Martin, 1984).
- Feferman, S.: 1964, 'Systems of Predicative Analysis', *Journal of Symbolic Logic* **29**, 1–30.
- Feferman, S.: 1991, 'Reflecting on Incompleteness', *Journal of Symbolic Logic* **56**, 1–49.
- Gabbay, D. and Günthner, F. (eds.): 1989, *Handbook of Philosophical Logic* vol. IV, Reidel, Dordrecht.
- Girard J.-Y.: 1987, *Proof Theory and Logical Complexity* vol. 1, Bibliopolis, Napoli.
- Halbach, V.: 1994, *Tarski-Hierarchien*, Dissertation, Centrum für Informations- und Sprachverarbeitung, München.
- Halbach, V.: 1996, 'Tarskian and Kripkean Truth', forthcoming in the *Journal of Philosophical Logic*.
- Jokusch, C. G.: 1975, 'Recursiveness of Initial Segments of Kleene's $\mathcal{O}$ ', *Fundamenta Mathematica* **LXXXVII**, 161–167.
- Kleene, S. C.: 1944, 'On the Forms of Predicates in the Theory of Constructive Ordinals', *American Journal of Mathematics* **66**, 41–58.
- Kleene, S. C.: 1959, 'Quantification of Number-Theoretic Functions', *Compositio Mathematica* **14**, 23–40.
- Kripke, S.: 1975, 'Outline of a Theory of Truth', *Journal of Philosophy*, **72**, 690–716, reprinted in (Martin, 1984).
- Martin, R. L.: 1975, 'On Representing "true-in-L" in L', *Philosophia* **5**, 213–217, reprinted in (Martin, 1984).
- Martin, R. L. (ed.): 1984, *Recent Essays on Truth and the Liar Paradox*, Oxford University Press, Oxford.
- McCarthy, T.: 1988, 'Ungroundedness in Classical Languages', *Journal of Philosophical Logic* **17**, 61–74.
- McGee, V.: 1991, *Truth, Vagueness and Paradox*, Hackett Publishing, Indianapolis.
- Moschovakis, Y. N.: 1974, *Elementary Induction on Abstract Structures*, North Holland, Amsterdam.
- Parsons, C.: 1974, 'The Liar Paradox', *Journal of Philosophical Logic* **3**, 381–412, reprinted in (Martin, 1984).
- Rogers, H.: 1967, *Theory of Recursive Functions and Effective Computability*, McGraw-Hill Book Company, New York.

- Sacks, G.: 1990 *Higher Recursion Theory*, Springer, Berlin.
- Takeuti, G.: 1987, *Proof Theory*, second edition, North Holland, Amsterdam.
- Tarski, A.: 1935, 'Der Wahrheitsbegriff in den formalisierten Sprachen', *Studia Philosophica Commentarii Societatis philosophicae Polonorum* 1, reprinted in (Tarski, 1986).
- Tarski, A.: 1986, *Collected Papers* vol. 1, Givant, S. and McKenzie, R. (eds.), Birkhäuser, Basel.
- Visser, A.: 'Semantics and the Liar Paradox', in (Gabbay/Günthner, 1989), 617–706.

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